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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत

Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

B.Tech II Sem I Mid Semester Examination, 25th Feb, 2025

Mid Sem-I

Subject: Mathematics-II

Course Code: MAL-201

Branch: IT

Maximum Marks. -15 marks

Time- 60 min.

Note: All questions are compulsory.

Q-1	Find the volume of the portion of the sphere $x^2 + y^2 + z^2 = b^2$ lying inside the cylinder $x^2 + y^2 = bx$ by transforming to cylindrical coordinates.	(3marks)
Q-2	Evaluate the following integral by changing the order of integration $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2 dy dx.$	(3marks)
Q-3	Solve: $\left(x - \frac{y}{x^2+y^2}\right) dx + \left(y + \frac{x}{x^2+y^2}\right) dy = 0.$	(3marks)
Q-4	Solve: $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y.$	(3marks)
Q-5	Find the solution of initial value problem $y' = 2y - x$ with $y(0) = 1$, By using Picard's Iteration Method upto 5 th Iteration. Also compare with exact solution.	(3marks)



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(An Autonomous Institute of National Importance under Act of Parliament)

Phone: +91 1302987902, Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

Mid Sem-I

Subject Name: Data Structure

Subject code: UIT 203

Roll no. 12412003

Branch: IT

2nd Semester

M.M.-15 marks

Time- 60 min.

Note: (1) Attempt all the questions.

(2) All questions carry equal marks (5*3=15)

Q-1	What is sorting in data structure? Explain the sorting as given below with the help of suitable example. i) Bubble sort ii) Merge Sort iii) Quicksort	(5 marks)
Q-2	Analyze the time complexity of various array (n length) operations such as a. Accessing an element, b. Searching for an element c. Finding minimum and maximum d. Sorting an array. e. Reversing an Array f. Finding Duplicate elements in array g. Union/Intersection of Two Arrays	(5 marks)
Q-3	What are the various Operations performed on the Stack? Write an algorithm for matching different parenthesis in an algebraic expression. Compare queue and stack and justify in between these two which one is the more efficient. Write the time complexity of insertion, deletion and search operation in stack and queue.	(5 marks)

End



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2nd Semester Mid Sem-I

Subject: Web Designing (UCS 003)

Roll no: 12412003

Branch: CSE/IT

M.M.-15 marks

Time - 60 minutes

Note: All questions are compulsory.

Q-1	Define CSS. Explain inline, internal, external and embedded style sheets with suitable examples and possible outcomes also. List in bullet points the advantages and disadvantages of each.	(3 M)																
Q-2	Create a webform including 5 tags. Additionally, create a drop down box, numeric slider. How do we make sure that form data is never saved automatically in the browser?	(3 M)																
Q-3	Show how merge and alignment of tables, rows and columns is achieved using HTML with suitable examples? Explain the given format and write HTML code for the same. <table><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td colspan="2"></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></table>																	(3 M)
Q-4	Write briefly: a. Describe any 2 basic formatting tags b. How to do external linking in HTML document c. Insert an image d. What is the difference between COLSPAN and ROWSPAN? e. Create an ordered list f. Create rounded/curved element	(3 M)																
Q-5	How to create an animation in HTML5. Explain both shorthand and detailed notations along with how to set up keyframes.	(3 M)																



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Mid Sem-I

Branch: IT/CSE/DS
(2nd SEM)

Subject: OOPs using C++ (UCS004)
M.M.-15 marks

Roll no. 12412003

Time- 60 mint.

Note: All questions are compulsory.

Question No.	Questions	Marks
Q-1: (a)	How does encapsulation improve code maintainability and security in OOP?	2
(b)	Write a code snippet to demonstrate how to access array elements using pointers in C++.	2
Q-2:	Provide an example of a real-world problem and explain how you would apply object-oriented design principles to solve it.	3
Q-3:	Critically examine how scope resolution operator (::) improves code clarity and modularity in C++ programs.	4
Q-4: (a)	Explain the concept of inline functions. When should they be used, and what are their limitations?	2
(b)	What are arrays of pointers and pointers to pointers? Provide an example of how they can be used in a program.	2



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Mid Sem-I
Branch: CSE/IT

Subject: Digital Electronics (ECL-202)
M.M.-15 marks

Roll no. 12412003
Time- 60 mint.

Note: All questions are compulsory.

Q-1	Convert the decimal number $(341)_{10}$ to binary, octal and base-4 number system.	(3 marks)
Q-2	Simplify the below function using Boolean's law and draw the circuit diagram for the minimized expression. $f(A, B, C) = ABC + AB'C' + AB'C + ABC' + A'B'C$	(3 marks)
Q-3	Compare and explain radix complement and diminished radix complement in the binary number system with examples.	(3 marks)
Q-4	Explain the process of BCD addition and simplify the expression $146 + 75$ using BCD Addition.	(3 marks)
Q-5	Minimize the following function using K-Map and draw a circuit diagram for the minimized expression using only NAND gates. $F = \sum (0, 1, 2, 5, 7, 9, 13) + d(8, 10, 15)$	(3 marks)



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COMMUNICATION SKILLS

M.M.15 Marks

MID SEM-1

SEM-II

Batch-2024-28

Time-60 min.

Attempt All. (3 X 5 = 15)

Roll No.

1 2 4 1 2 0 0 3

1. What do you understand by 'non-verbal communication'? Discuss how non-verbal communication plays a significant role in contemporary modern organizations. Elaborate with suitable examples.
2. Write a detailed essay on Flow of communication. Write the advantages and disadvantages of each type of flow of communication.
3. What do you understand by 'barriers to communication'? With the help of suitable examples, discuss its various types.